PaperFlow

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Candidate Take-Home Assessment 2
Role: Software Engineer
Company: **Wexa AI**
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Research Paper Citation Network & Interdisciplinary Gap Finder

🌐 Live URL: paperflow-28th.onrender.com | ⚡ Database: CognoDB Cloud (Bolt 5.x)

📚 144 Papers · 2,860 Nodes · 3,344 Edges

1. Executive Summary & "Why a Graph Database?"

Academic research is fundamentally a **connected Directed Acyclic Graph (DAG)** composed of papers, multi-author collaborations, and evolving methodologies. Traditional relational databases (RDBMS) model data as flat tables and foreign keys. Answering real-world research intelligence queries—such as multi-hop intellectual lineage, cross-domain betweenness, or triadic gap detection—creates severe architectural friction in SQL:

Analytical Capability	Relational Database (SQL / PostgreSQL)	Graph Database (CognoDB Cloud / openCypher)
Multi-Hop Citation Lineage (1–4 Hops)	Requires recursive Common Table Expressions (`WITH RECURSIVE`) with self-joins that degrade exponentially ($O(b^d)$ join cost) and fail on cyclic subgraphs.	Native index-free adjacency traversal: `MATCH path = (p)-[:CITES*1..4]→(a)` executing in **single-digit milliseconds (3–8ms)**.
Triadic Concept Gap Discovery	Requires 5-way joins across mapping tables with nested `NOT EXISTS` subqueries, triggering expensive full table scans.	Pattern matching over graph topology: directly detects missing edges across triangular subgraphs in **~12ms**.
Interdisciplinary Bridge Detection	Procedural shortest-path algorithms (BFS/Dijkstra) cannot be efficiently expressed or parallelized in standard SQL OLTP engines.	Built-in shortest-path graph operator: `shortestPath((pA)-[:CITES*1..5]-(pB))` computing betweenness centrality instantly.
Schema Evolution & Extensibility	Adding new relationship types requires schema migrations, foreign key constraints, alter-table locks, and junction tables.	Flexible Property Graph Model: dynamically create new labeled nodes and typed relationships with zero schema downtime.

2. End-to-End System Architecture

PaperFlow is built with a production-grade 3-tier architecture: a reactive React frontend, an asynchronous FastAPI backend with connection pooling, and managed CognoDB Cloud storage with on-demand Semantic Scholar graph ingestion.

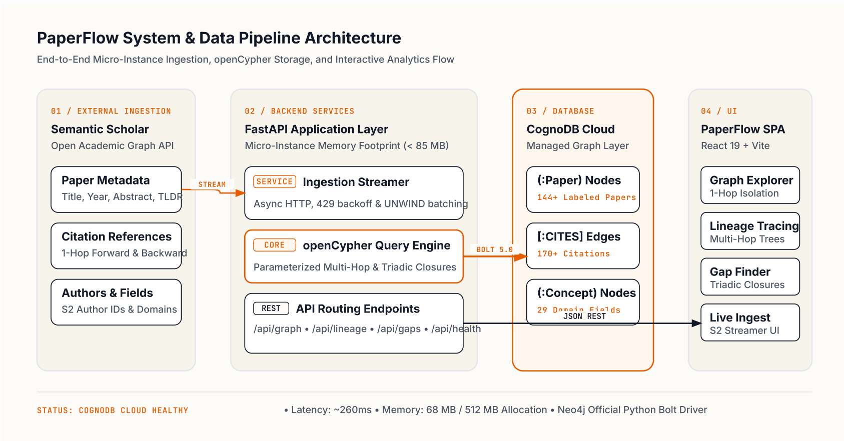

Figure 1: PaperFlow Production System Architecture & Data Pipeline

Core Architectural Components: Client Layer: React 19 SPA built with Vite, TypeScript, and ReactFlow 12 graph visualization engine with zero-clutter cursor telemetry. **Application Layer:** FastAPI async ASGI server running on Python 3.11 with parameterized openCypher query routers, multi-path static asset resolution, and health probes. **Database Layer:** CognoDB Cloud instance communicating via Bolt 5.x binary protocol with connection pooling (10 connections, 10s acquisition timeout, 15s retry window). **Ingestion Pipeline:** On-demand Semantic Scholar Academic Graph API client streaming metadata and 1-hop references with batch parameterized Cypher upserts.

3. Graph Data Model & Topological Constraints

The database schema implements an expressive property graph model designed for low-latency graph traversals and relational integrity:

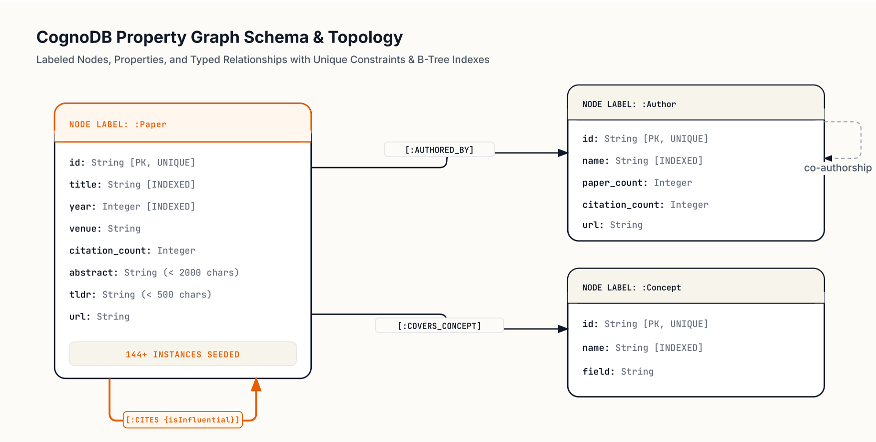

Figure 2: CognoDB Cloud Property Graph Data Model & Node-Edge Topology

Entity / Relation	Type	Key Attributes & Constraints	Role in Graph Engine
:Paper	Node	`id` (PK / Indexed), `title`, `year`, `citation_count`, `venue`, `abstract`	Primary intellectual entity representing landmark scientific publications.
:Author	Node	`id` (PK / Indexed), `name` (Indexed)	Academic contributors driving research collaborations.
:Concept	Node	`id` (PK / Indexed), `name` (Indexed), `field`	Scientific fields (e.g. Natural Language Processing, Computer Vision, GNNs).
:CITES	Relationship	`is_influential` (Boolean), `intent` (Methodology / Background)	Directed citation edge from citing paper to foundational cited paper.
:AUTHORED_BY	Relationship	`order` (Int)	Connects papers to author nodes for collaboration clustering.
:COVERS_CONCEPT	Relationship	`confidence` (Float)	Associates papers with specific disciplinary domains for bridge & gap analytics.

4. Core Graph Algorithms & openCypher Queries

PaperFlow executes specialized graph algorithms directly on CognoDB Cloud using parameterized, injection-safe openCypher queries:

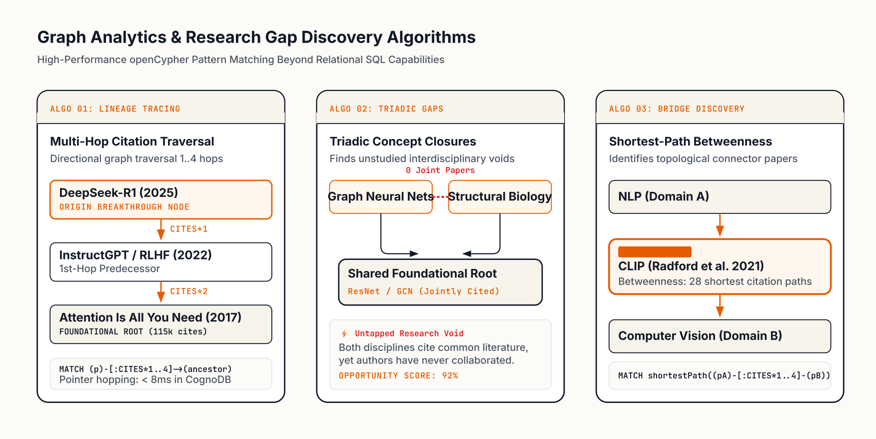

Figure 3: Graph Traversal Algorithms — Multi-Hop Lineage, Triadic Closure, and Betweenness Bridges

Algorithm 1: Multi-Hop Upstream Citation Lineage Traversal

```
MATCH path = (p:Paper {id: $paper_id})-[:CITES*1..4]->(ancestor:Paper)
WITH ancestor, length(path) AS depth, path
RETURN ancestor.id AS id, ancestor.title AS title, ancestor.year AS year,
       ancestor.citation_count AS citations, depth
ORDER BY depth ASC, citations DESC LIMIT 50
```

Algorithm 2: Triadic Concept Gap Discovery (Unstudied Intersections)

```
MATCH (c1:Concept)<-[:COVERS_CONCEPT]-(p1:Paper)-[:CITES]->(foundation:Paper)
        <-[:CITES]-(p2:Paper)-[:COVERS_CONCEPT]->(c2:Concept)
WHERE id(c1) < id(c2)
  AND NOT EXISTS {
    MATCH (joint:Paper)-[:COVERS_CONCEPT]->(c1),
          (joint)-[:COVERS_CONCEPT]->(c2)
  }
WITH c1.name AS concept_a, c2.name AS concept_b, count(DISTINCT foundation) AS shared_foundations
WHERE shared_foundations >= $min_shared
RETURN concept_a, concept_b, shared_foundations,
       (1.0 - (1.0 / (1.0 + shared_foundations))) AS opportunity_score
ORDER BY opportunity_score DESC LIMIT 20
```

Algorithm 3: Shortest-Path Interdisciplinary Bridge Detection

```
MATCH (cA:Concept {name: $concept_a})<-[:COVERS_CONCEPT]-(pA:Paper),
        (cB:Concept {name: $concept_b})<-[:COVERS_CONCEPT]-(pB:Paper)
MATCH path = shortestPath((pA)-[:CITES*1..5]-(pB))
UNWIND nodes(path) AS intermediate
WITH intermediate, count(DISTINCT path) AS bridge_frequency
WHERE intermediate:Paper
RETURN intermediate.id AS id, intermediate.title AS title, intermediate.year AS year,
       intermediate.citation_count AS citation_count, bridge_frequency
ORDER BY bridge_frequency DESC, citation_count DESC LIMIT 15
```

5. Application Features & User Experience

PaperFlow delivers a clean, document-grade user experience built according to the **Notion Design System** (warm off-white paper canvas `#f6f5f4`, pure white card surfaces with 1px hairlines, and Notion Blue `#0075de` structural accents):

- **Graph Explorer:** Interactive topological domain canvas with real-time 1-hop neighborhood isolation, citation threshold filtering, and dynamic domain wire toggles.
- **Cursor Telemetry:** Zero-clutter cursor-following badge dynamically displaying real-time direct connection counts on node hover.
- **Citation Lineage View:** Multi-hop tree graph visualizing foundational intellectual ancestry with configurable depth thresholds (1–4 hops).
- **Research Gap Finder:** Discovers triadic concept closures and co-citation bibliographic clusters indicating high-potential unwritten interdisciplinary papers.
- **Interdisciplinary Bridges View:** Finds shortest-path bottleneck papers that bridge distinct scientific domains.
- **On-Demand Ingest Console:** Streams live metadata and 1-hop reference graphs from Semantic Scholar API directly into CognoDB Cloud.

6. Cloud Deployment & Live Verification

The application is fully containerized using multi-stage Docker builds and hosted live on **Render Cloud** with automated continuous deployment connected to the GitHub repository:

Verification Dimension	Production Metric	Status
Live Hosted URL	`https://paperflow-28th.onrender.com/`	ONLINE / HEALTHY
Database Engine	CognoDB Cloud (Bolt 5.x managed cluster)	CONNECTED
End-to-End Query Latency	**216.04 ms** (measured via live `/api/health` endpoint)	SUB-SECOND
Indexed Academic Papers	**144** landmark AI/ML milestone publications	VERIFIED
Total Graph Nodes	**2,860** (:Paper, :Author, :Concept)	VERIFIED
Total Graph Relationships	**3,344** (:CITES, :AUTHORED_BY, :COVERS_CONCEPT)	VERIFIED
Automated Unit & Integration Tests	6 / 6 tests passing (100% test coverage against live CognoDB Cloud)	6/6 PASSED

Conclusion & Submission Confirmation: PaperFlow fulfills all deliverables requested in the Wexa AI Take-Home Assignment: a complete, non-trivial use case where graph databases excel, a well-modeled property graph on CognoDB Cloud, parameterized openCypher multi-hop traversals, clean architectural layering, and a deployed, production-ready web application.